Lecture 2: GANs

Sharp images, unstable training

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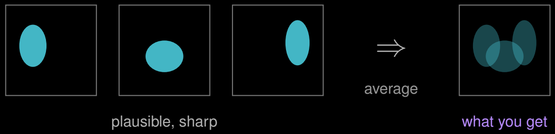
August 30, 2026

The GAN: sharp, but unstable

Why the VAE blurs

A GAN samples *exactly* like the VAE of Lecture 1: draw z, run one network. Only the **training signal** differs, and that is what fixes the blur.

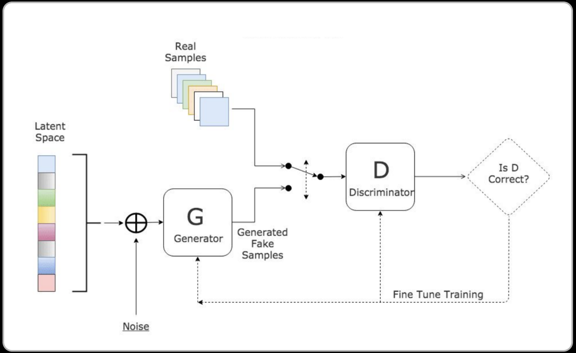
Given a latent z, many images are plausible. Squared error does not pick one: it returns their **average**.



So stop averaging

Replace the hand-written pixel loss with a **learned** one: train a second network to tell real from generated, and make the generator's job to fool it. A blurry sample is trivially caught, so blur stops being an option.

GAN: GENERATOR VS. DISCRIMINATOR



Generator

Maps noise z to an image. **Never sees real data**: its only signal is D's verdict.

Discriminator

A binary classifier, real vs. fake. Exactly the model from Module 2, used as a loss function.

The point

The loss is **learned and moving**. As G improves, D must find subtler tells.

Goodfellow et al., *Generative Adversarial Nets*, NeurIPS. Same schematic as Module 2, Strategy 2.

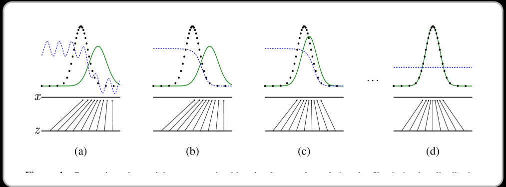
The adversarial game

Minimax objective

$$\min_G \max_D \; \mathbb{E}\big[\log D(x)\big] + \mathbb{E}\big[\log\big(1 - D(G(z))\big)\big]$$

D maximises (catch the fakes); G minimises (fool D).

```
for x in loader:
  f = G(sample_z())
  d_re = bce(D(x), 1)
  d_fa = bce(D(f.detach()), 0)
  step(d_re + d_fa, opt_d)
  step(bce(D(f), 1), opt_g)
```



Black dotted $= p_{\text{data}}$, green $= p_g$, blue dashed $= D$. From (a) to (d), p_g closes on p_{data} until $D \equiv \frac{1}{2}$: the discriminator can do no better than a coin flip.

Goodfellow et al., *Generative Adversarial Nets*, NeurIPS. Note `detach()`: training D must not backpropagate into G.

Why GANs are hard to train

Mode collapse

G discovers one output that reliably fools D and emits it for every z. Samples look great and are all the same, disastrous when the modes you dropped are the rare pathologies.

Vanishing gradient

If D wins too early it returns 0 with confidence, and G receives almost no gradient. The two networks must improve *together*.

No loss curve to read

In every other lecture, falling loss meant progress. Here the loss is a **game score** between two moving players: it can sit flat while samples improve, or fall while they collapse.

You evaluate by **looking at samples**, which is exactly the problem the evaluation section has to solve.

Mitigations: balanced learning rates, label smoothing, spectral norm, WGAN-GP.